PHYS1408 Practice Exam 2-1 — Answer Key

Worked solutions

All answers were checked numerically with `_scripts/verify_2_1.py`. $g = 9.8$ m/s^2 throughout.

Q	Answer	Value	Topic (source)
1	**a**	4.8 m	Work–energy with friction, halfpipe (Exam 2 #1, Ch 7/8 notes p. 4–5)
2	**c**	13.0 m	Projectile max height, review (Exam 2 #2)
3	**d**	17.1 m/s	Loop-the-loop, minimum entry speed $v_{bottom} = \sqrt{5gR}$ (Exam 2 #3, Ch 7/8 notes p. 12/18–13/18)
4	**b**	0.98 s	Sliding up a rough ramp, review (Exam 2 #4)
5	**d**	20.9 m/s	Banked curve v_{max} (Ch 6 notes p. 3–5)
6	**c**	1.2 m	Vertical spring launch (Exam 2 #5, Ch 7/8 notes p. 12/18)
7	**d**	-1.0 m/s	1-D elastic collision (Exam 2 #6, Ch 9 notes p. 2–3)
8	**c**	30–40%	Perfectly inelastic, KE fraction (Exam 2 #7, Ch 9 notes p. 1)
9	**e**	314 m/s	Ballistic pendulum (Exam 2 #8, Ch 9 notes p. 5)
10	**b**	3.4 m/s	Hanging mass on disk pulley (Exam 2 #9, Ch 10 notes p. 4–6)
11	**c**	102 rad/s	Rolling without slipping (Exam 2 #10, Ch 10 notes p. 6–8)

Q	Answer	Value	Topic (source)
12	**a**	+1.09 rad/s	Angular-momentum "collision" (Exam 2 #11, Ch 11 notes p. 1, 3)
13	**b**	0.82 rad/s	Running jump onto merry-go-round (Exam 2 #12, Ch 11 notes p. 4)
14	**e**	2.1 rad/s	Walking to the center (Ch 11 notes p. 7)
15	**a**	2.08 m/s	Incline + friction + pulley, energy (Exam 2 #13, Ch 7/8 notes p. 13–16)

1. Halfpipe with a rough patch — a) 4.8 m

Work–energy theorem, $E_i = E_f + E_{lost}$. Energy is lost only on the rough patch. She starts and ends at rest, so only PE appears.

Energy terms:

$$E_i = mgh_i$$

$$E_f = mgh_f$$

$$E_{lost} = F_f L = \mu N L = \mu mgL$$

Solve for h_f:

$$mgh_i = mgh_f + \mu mgL$$

$$h_i = h_f + \mu L$$

$$h_f = h_i - \mu L$$

Plug in:

$$h_f = 6.0 - (0.15)(8.0) = 6.0 - 1.2 = 4.8 \text{ m}$$

Mass cancels. Distractor d) 6.0 m is the frictionless answer, $h_f = h_i$.

2. Projectile from a ledge — c) 13.0 m

Only the vertical motion decides the height. At the top the vertical velocity is zero. Take up as +, so the vertical acceleration is $-g$ (worked example: Worked Problems R.1).

Vertical launch speed:

$$v_{0y} = v_0 \sin\theta$$

Vertical kinematics, launch point to the top ($v_y = 0$):

$$v_y^2 = v_{0y}^2 - 2g\,\Delta y$$

$$0 = (v_0 \sin\theta)^2 - 2g\,\Delta y$$

$$\Delta y = \frac{(v_0 \sin\theta)^2}{2g}$$

Add the ledge:

$$H = h + \frac{(v_0 \sin\theta)^2}{2g}$$

Plug in:

$$H = 5.0 + \frac{(25 \sin 30^\circ)^2}{2(9.8)}$$

$$= 5.0 + \frac{(12.5)^2}{19.6} = 5.0 + 7.97 = 13.0 \text{ m}$$

Distractors:

- a) 5.0 m is just the ledge.
- b) 8.0 m forgets to add the ledge.
- d) 21.0 m forgets the 2 in $2g$: $5.0 + (v_0 \sin\theta)^2/g = 5.0 + 15.9$.
- e) 36.9 m is $h + v_0^2/2g$, as if launched straight up.

3. Loop-the-loop — d) 17.1 m/s

Top of the loop (forces). To "barely" stay on, $N = 0$ at the top, so gravity alone supplies the centripetal force:

$$mg = \frac{mv_{top}^2}{R}$$

$$v_{top}^2 = gR$$

Bottom to top (energy), $E_i = E_f$. Put $h = 0$ at the bottom; the top is at $2R$.

$$E_i = \tfrac{1}{2}mv_b^2$$

$$E_f = \tfrac{1}{2}mv_{top}^2 + mg(2R)$$

Solve for v_b:

$$\tfrac{1}{2}mv_b^2 = \tfrac{1}{2}mv_{top}^2 + 2mgR$$

$$v_b^2 = v_{top}^2 + 4gR = gR + 4gR = 5gR$$

$$v_b = \sqrt{5gR}$$

Plug in:

$$v_b = \sqrt{5(9.8)(6.0)} = \sqrt{294} = 17.1 \text{ m/s}$$

Distractor a) 7.7 m/s is $\sqrt{gR}$, the speed needed at the *top*.

4. Sliding up a rough ramp — b) 0.98 s

Going up, both gravity's component and kinetic friction point down the slope:

$$a = g(\sin\theta + \mu_k \cos\theta) = 9.8(\sin 20^\circ + 0.30\cos 20^\circ) = 9.8(0.342 + 0.282) = 6.11 \text{ m/s}^2 \text{ (decelerating)}.$$

$$v_f = v_i - at = 0 \quad \Rightarrow \quad t = \frac{v_i}{a} = \frac{6.0}{6.11} = 0.98 \text{ s}.$$

The 3.0 kg mass cancels. Distractor d) 1.8 s ignores friction ($a = g\sin\theta$).

5. Banked curve, maximum speed — d) 20.9 m/s

At v_{max} the car tends to slide *up* the bank, so static friction points *down* the bank. Summing forces (the professor's derivation, Ch 6 p. 3–4):

$$v_{max} = \sqrt{gr\,\frac{\mu_s + \tan\theta}{1 - \mu_s \tan\theta}}.$$

With $\tan 15^\circ = 0.268$:

$$v_{max} = \sqrt{9.8(80)\,\frac{0.25 + 0.268}{1 - (0.25)(0.268)}} = \sqrt{784\,\frac{0.518}{0.933}} = \sqrt{435} = 20.9 \text{ m/s}.$$

Distractors: a) $\sqrt{\mu_s gr} = 14.0$ (flat curve), b) $\sqrt{gr\tan\theta} = 14.5$ (frictionless bank).

6. Vertical spring launch — c) 1.2 m

$E_i = E_f$. Put $h = 0$ at the release point. Spring PE becomes gravitational PE at the top, where $v = 0$.

Energy terms:

$$E_i = \tfrac{1}{2}kx^2$$

$$E_f = mgh$$

Solve for h:

$$\tfrac{1}{2}kx^2 = mgh$$

$$h = \frac{kx^2}{2mg}$$

Plug in:

$$h = \frac{1500(0.040)^2}{2(0.10)(9.8)} = \frac{2.4}{1.96} = 1.2 \text{ m}$$

Distractor d) 2.4 m forgets the $\frac{1}{2}$.

7. 1-D elastic collision — d) −1.0 m/s

Use the notecard formulas from Ch 9 (p. 3), with right positive: $m_1 = 3.0$, $v_{1i} = +4.0$, $m_2 = 1.0$, $v_{2i} = -6.0$.

$$v_{1f} = \frac{(m_1 - m_2)v_{1i} + 2m_2 v_{2i}}{m_1 + m_2} = \frac{(2)(4.0) + 2(1)(-6.0)}{4.0} = \frac{8 - 12}{4} = -1.0 \text{ m/s}.$$

Check: $v_{2f} = v_{1i} + v_{1f} - v_{2i} = 4 - 1 + 6 = +9.0$ m/s. Momentum: $3(4) + 1(-6) = 6$ before, and $3(-1) + 1(9) = 6$ after. (checks) KE: $24 + 18 = 42$ J before, and $1.5 + 40.5 = 42$ J after. (checks)

The 3 kg ball bounces back to the left at 1.0 m/s. Distractor a) +9.0 m/s is the 1 kg ball's final velocity.

8. Perfectly inelastic, KE fraction — c) 30–40%

Momentum is conserved: $m(3v) + 2m(0) = (3m)v_f$, so $v_f = v$.

$$K_i = \tfrac{1}{2}m(3v)^2 = \tfrac{9}{2}mv^2, \qquad K_f = \tfrac{1}{2}(3m)v^2 = \tfrac{3}{2}mv^2.$$

$$\frac{K_f}{K_i} = \frac{3/2}{9/2} = \frac{1}{3} \approx 33\%.$$

9. Ballistic pendulum — e) 314 m/s

There are two stages (Ch 9 p. 5).

1. **Swing (energy conserved):** $\frac{1}{2}(M+m)V^2 = (M+m)gh$, so $V = \sqrt{2gh} = \sqrt{2(9.8)(0.080)} = 1.25$ m/s.
2. **Collision (momentum conserved):** $mv = (M+m)V$, so

$$v = \frac{M+m}{m}\sqrt{2gh} = \frac{2.008}{0.0080}(1.25) = 251 \times 1.25 = 314 \text{ m/s}.$$

Do **not** use energy conservation across the collision; most of the KE is lost there. Distractor a) 1.25 m/s is V itself, and d) 251 is just the mass ratio.

10. Hanging mass on a disk pulley — b) 3.4 m/s

Energy route, $E_i = E_f$. Put $h = 0$ where the mass ends up. It starts from rest, and the pulley is a disk with $I = \frac{1}{2}MR^2$ turning at $\omega = v/R$.

Energy terms:

$$E_i = mgh$$

$$E_f = \tfrac{1}{2}mv^2 + \tfrac{1}{2}I\omega^2$$

Pulley KE in terms of v:

$$\tfrac{1}{2}I\omega^2 = \tfrac{1}{2}\left(\tfrac{1}{2}MR^2\right)\left(\frac{v}{R}\right)^2 = \tfrac{1}{4}Mv^2$$

Solve for v:

$$mgh = \tfrac{1}{2}mv^2 + \tfrac{1}{4}Mv^2 = \tfrac{1}{2}\left(m + \tfrac{M}{2}\right)v^2$$

$$v^2 = \frac{2mgh}{m + \frac{M}{2}}$$

$$v = \sqrt{\frac{2mgh}{m + \frac{M}{2}}}$$

Plug in:

$$v = \sqrt{\frac{2(2.0)(9.8)(1.5)}{2.0 + 3.0}} = \sqrt{11.76} = 3.43 \text{ m/s}$$

Torque route (check). From $RT = I\alpha$, $T = \frac{1}{2}Ma$. From $mg - T = ma$:

$$a = \frac{mg}{m + \frac{M}{2}} = \frac{19.6}{5.0} = 3.92 \text{ m/s}^2$$

$$v = \sqrt{2ah} = \sqrt{2(3.92)(1.5)} = 3.43 \text{ m/s (checks)}$$

R cancels. Distractor d) 5.4 m/s is free fall, $\sqrt{2gh}$.

11. Rolling solid cylinder — c) 102 rad/s

$E_i = E_f$. Put $h = 0$ at the bottom. Rolling without slipping means $v = R\omega$, and a solid cylinder has $I = \frac{1}{2}mR^2$.

Energy terms:

$$E_i = mgh$$

$$E_f = \tfrac{1}{2}mv^2 + \tfrac{1}{2}I\omega^2$$

Solve for v, then ω:

$$mgh = \tfrac{1}{2}mv^2 + \tfrac{1}{2}\left(\tfrac{1}{2}mR^2\right)\frac{v^2}{R^2} = \tfrac{1}{2}mv^2 + \tfrac{1}{4}mv^2 = \tfrac{3}{4}mv^2$$

$$v = \sqrt{\tfrac{4}{3}gh}$$

$$\omega = \frac{v}{R} = \frac{1}{R}\sqrt{\tfrac{4}{3}gh}$$

Plug in:

$$v = \sqrt{\tfrac{4}{3}(9.8)(2.0)} = 5.11 \text{ m/s}$$

$$\omega = \frac{5.11}{0.050} = 102 \text{ rad/s}$$

Distractors:

- a) 5.1 is v, not ω.
- b) 89 uses the ring formula, $\sqrt{gh}$.
- d) 125 uses the sliding formula, $\sqrt{2gh}$.

12. Counter-rotating ring dropped on disk — a) +1.09 rad/s

$$I_{disk} = \tfrac{1}{2}(300)(2.0)^2 = 600 \text{ kg}\cdot\text{m}^2, \qquad I_{ring} = (100)(1.5)^2 = 225 \text{ kg}\cdot\text{m}^2.$$

$$L_i = L_f: \quad \omega_f = \frac{I_{disk}\,\omega_{disk} + I_{ring}\,\omega_{ring}}{I_{disk} + I_{ring}} = \frac{600(3.0) + 225(-4.0)}{825} = \frac{1800 - 900}{825} = +1.09 \text{ rad/s}.$$

The sign matters: e) +3.27 is what you get by forgetting the minus sign on the ring.

13. Child jumps onto the rim — b) 0.82 rad/s

The child's angular momentum about the axle before landing is $L = mvR$ (Ch 11 p. 4). After landing, the child is a point mass at R.

$$\omega_f = \frac{\frac{1}{2}MR^2\omega_i + mvR}{\frac{1}{2}MR^2 + mR^2} = \frac{\frac{1}{2}(400)(2.5)^2(0.50) + 40(6.0)(2.5)}{\frac{1}{2}(400)(2.5)^2 + 40(2.5)^2} = \frac{625 + 600}{1250 + 250} = \frac{1225}{1500} = 0.82 \text{ rad/s}.$$

Distractors: a) 0.42 ignores the child's running momentum; c) 0.98 forgets to add the child's mR^2 to the final I.

14. Walking from the rim to the center — e) 2.1 rad/s

No external torque, so $L_i = L_f$. At the center the person contributes $md^2 = 0$.

$$\left(\tfrac{1}{2}MR^2 + mR^2\right)\omega_i = \tfrac{1}{2}MR^2\,\omega_f \quad \Rightarrow \quad \omega_f = \frac{M + 2m}{M}\,\omega_i = \frac{400 + 160}{400}(1.5) = 1.4(1.5) = 2.1 \text{ rad/s}.$$

The merry-go-round speeds up, and its KE *increases* (the person does work walking inward; Ch 11 p. 8).

15. Incline + friction + disk pulley — a) 2.08 m/s

Work–energy theorem, $E_i = E_f + E_{lost}$.

- Put $h = 0$ at the block's final position, so it starts $d\sin\theta$ higher.
- The pulley is a disk, $I = \frac{1}{2}MR^2$, turning at $\omega = v/R$.
- On the incline, $N = mg\cos\theta$.

Energy terms:

$$E_i = \tfrac{1}{2}mv_i^2 + \tfrac{1}{2}I\omega_i^2 + mgd\sin\theta$$

$$E_f = \tfrac{1}{2}mv_f^2 + \tfrac{1}{2}I\omega_f^2$$

$$E_{lost} = \mu_k mg\cos\theta\, d$$

Pulley KE in terms of v:

$$\tfrac{1}{2}I\omega^2 = \tfrac{1}{2}\left(\tfrac{1}{2}MR^2\right)\left(\frac{v}{R}\right)^2 = \tfrac{1}{4}Mv^2$$

Substitute and solve for v_f^2:

$$\tfrac{1}{2}mv_i^2 + \tfrac{1}{4}Mv_i^2 + mgd\sin\theta = \tfrac{1}{2}mv_f^2 + \tfrac{1}{4}Mv_f^2 + \mu_k mg\cos\theta\, d$$

$$\tfrac{1}{2}\left(m + \tfrac{M}{2}\right)v_f^2 = \tfrac{1}{2}\left(m + \tfrac{M}{2}\right)v_i^2 + mgd(\sin\theta - \mu_k\cos\theta)$$

$$v_f^2 = v_i^2 + \frac{2mgd(\sin\theta - \mu_k\cos\theta)}{m + \frac{M}{2}}$$

Take the square root:

$$v_f = \sqrt{v_i^2 + \frac{2mgd(\sin\theta - \mu_k\cos\theta)}{m + \frac{M}{2}}}$$

Plug in:

$$v_f = \sqrt{0.50^2 + \frac{2(4.0)(9.8)(1.0)(0.4226 - 0.15\times 0.9063)}{4.0 + \frac{3.0}{2}}}$$

$$= \sqrt{0.25 + \frac{22.48}{5.5}} = \sqrt{0.25 + 4.087} = \sqrt{4.337} = 2.08 \text{ m/s}$$

The bracket $(\sin\theta - \mu_k\cos\theta)$ is positive, so the block speeds up and does not stop, which rules out e).

Distractors:

- b) 2.42 ignores the pulley.
- c) 2.50 ignores friction.
- d) 2.92 ignores both.